

$T_3:$

$$\frac{\partial z}{\partial x} = \frac{1}{\sqrt{x} + \sqrt{y}} \cdot \left(-\frac{1}{2\sqrt{x}} + 0 \right) =$$

$$= -\frac{1}{2\sqrt{x}(\sqrt{x} + \sqrt{y})}$$

$$\frac{\partial z}{\partial x} \in \mathbb{R}^{1 \times 1} \in \mathbb{R}^{1 \times 1}$$

由对称性

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{y}(\sqrt{x} + \sqrt{y})}$$

$$\frac{\partial z}{\partial y} \in \mathbb{R}^{1 \times 1} \in \mathbb{R}^{1 \times 1}$$

故:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = -\frac{\sqrt{x} + \sqrt{y}}{2(\sqrt{x} + \sqrt{y})} = -\frac{1}{2}$$

证毕

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{\sqrt{x} + \sqrt{y}}{2(\sqrt{x} + \sqrt{y})} = \frac{1}{2}$$

于是有

由假设

这一

在点

不论求号

前已

此,我

即

用数

T_5 :

$$(1) u = \sin(x^2 + y^2)$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = \cos(x^2 + y^2) \cdot (2x) dx + \cos(x^2 + y^2) \cdot (2y) dy$$

$$= 2 \cos(x^2 + y^2) (x dx + y dy)$$

(3)

$$u = e^{xy}$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = y \cdot e^{xy} dx + x \cdot e^{xy} dy$$

$$= e^{xy} (y dx + x dy)$$

(5)

这-
y) 在点 (

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = \cos(x^2+y^2) \cdot (2x) dx + \cos(x^2+y^2) \cdot (2y) dy$$

不论求导
前已

$$= 2 \cos(x^2+y^2) (x dx + y dy)$$

由此, 我们

d^2u , 即

$$u = e^{xy}$$

用数:

$$\begin{aligned} du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = y \cdot e^{xy} dx + x \cdot e^{xy} dy \\ &= e^{xy} (y dx + x dy) \end{aligned}$$

(5).

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

$$\begin{aligned} &= \frac{2x dx}{2\sqrt{x^2+y^2+z^2}} + \frac{2y dy}{2\sqrt{x^2+y^2+z^2}} + \frac{2z dz}{2\sqrt{x^2+y^2+z^2}} \\ &= \frac{x dx + y dy + z dz}{\sqrt{x^2+y^2+z^2}} \end{aligned}$$

(1) $z =$

(3) $z =$

(5) $u =$

设

设

$\frac{\partial x}{\partial y} = 2$

$$\Phi = f_{yx}(x + \theta_2 \Delta x, y + \theta_1 \Delta y) \Delta x \Delta y \quad (0 < \theta_1 < 1, 0 < \theta_2 < 1).$$

上面的 Φ 又可以

证:

(1)

$$\frac{\partial u}{\partial x} = 2x - 2y \Rightarrow \frac{\partial^2 u}{\partial x \partial y} = (2x - 2y)'_y = -2.$$

$$\frac{\partial u}{\partial y} = -6y - 2x \Rightarrow \frac{\partial^2 u}{\partial y \partial x} = (-6y - 2x)'_x = -2.$$

故 $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}.$

是有

假设 f_{yx} 及 f

这一定理

在点 (x, y)

不论求导数

(2)

在点 (x, y)

$$\frac{\partial u}{\partial x \partial y} = \frac{\partial u}{\partial y \partial x}.$$

不论求导数
前已定

由此, 我们

用数

(2) $u = x^y$

$$\frac{\partial u}{\partial x} = y \cdot x^{y-1} \Rightarrow \frac{\partial^2 u}{\partial x \partial y} = (y \cdot x^{y-1})'_y = y \cdot x^{y-1} + y \cdot \ln x \cdot x^{y-1}$$

$$= y \cdot x^{y-1} (2 + \ln x)$$

$$\frac{\partial u}{\partial y} = x \ln x \cdot x^{y^2} \quad \frac{\partial^2 u}{\partial y \partial x} = (\ln x \cdot x^{y^2})'_x = \frac{2y}{x} \cdot x^y + \ln x \cdot y^2 \cdot x^{y-1} \cdot y$$

$$= y \cdot x^{y^2-1} (2 + 2y^2 \ln x)$$

故: $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

(3)

不妨设 $y = x > 0$, 可得:

此时
微, 则

$$\frac{\partial u}{\partial x} = -\frac{1}{\sqrt{1-\frac{x}{y}}} \times \frac{1}{\sqrt{y}} \times \frac{1}{2\sqrt{x}} = -\frac{1}{2\sqrt{x(y-x)}}$$

及

这个

$$\frac{\partial u}{\partial y} = -\frac{1}{\sqrt{1-\frac{x}{y}}} \times \sqrt{x} \times (-\frac{1}{2}) \times \frac{1}{y^{\frac{3}{2}}} = \frac{\sqrt{x}}{2y\sqrt{y-x}}$$

由

$$\frac{\partial^2 u}{\partial x \partial y} = \left(\frac{1}{2\sqrt{xy-x^2}} \right)'_y = -\frac{x}{4(xy-x^2)^{\frac{3}{2}}}$$

由

$$\frac{\partial^2 u}{\partial y \partial x} = \left(\frac{\sqrt{x}}{2y\sqrt{y-x}} \right)'_x = -\frac{x}{4(xy-x^2)^{\frac{3}{2}}}$$

得证.

从而

$$\frac{\partial y}{\partial x}$$

$$(y\sqrt{y-x})^x$$

$$4(xy-x^2)^{\frac{1}{2}}$$

1812.

而定

1.

完全

(11).

$$\frac{\partial u}{\partial r} = \frac{\partial f}{\partial x} \cdot \left(\frac{\partial x}{\partial r}\right) + \frac{\partial f}{\partial y} \cdot \left(\frac{\partial y}{\partial r}\right) = \frac{\partial f}{\partial x} \cdot \cos \theta + \frac{\partial f}{\partial y} \cdot \sin \theta$$

则

$$\frac{\partial^2 u}{\partial r^2} = \frac{\partial^2 f}{\partial x^2} \cdot \cos^2 \theta + \frac{\partial^2 f}{\partial x \partial y} \sin 2\theta + \frac{\partial^2 f}{\partial y^2} \cdot \sin^2 \theta$$

那

等

通

而

(12)

$$\frac{\partial u}{\partial \xi} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \xi} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \xi} = \frac{\partial f}{\partial x} \cdot a + 0 = \frac{\partial f}{\partial x} \cdot a = af_x$$

$$\frac{\partial^2 u}{\partial \xi^2} = af_x \cdot af_x = a^2 f_{xx}$$

6. 若 $u = f(r)$,

$$\frac{\partial^2 u}{\partial x \partial y} = a f_x \cdot b f_y = ab f_{xy}.$$

7. 若 u, v 为

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial y} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial y} = b f_y.$$

明等式

8. 设 $f(tx, ty)$

$$\frac{\partial^2 u}{\partial y^2} = b f_y \cdot b f_y = b^2 f_{yy}.$$

具有这样性质

9. 设 $\varphi =$

(13).

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} = 2x f'(x^2 + y^2 + z^2)$$

具有这样性质
9) 设 φ

(13).

满足

10. 设

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} = 2x f'(x^2 + y^2 + z^2)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} (2x f'(x^2 + y^2 + z^2)) = 2 f'(x^2 + y^2 + z^2) + 4x^2 \cdot f''(x^2 + y^2 + z^2)$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial y} (2x f'(x^2 + y^2 + z^2)) = 2x \cdot 2y \cdot f''(x^2 + y^2 + z^2) = 4xy \cdot f''(x^2 + y^2 + z^2)$$

$$\frac{\partial u}{\partial y} = 2y f'(x^2 + y^2 + z^2)$$

$$\frac{\partial u}{\partial z} = 2z f'(x^2 + y^2 + z^2)$$

6. 若 $u = f(r)$, $r = \sqrt{x^2 + y^2}$, 其中 $f(r)$ 二次可微, 试证明 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr}$.

(14) $u = f(x, y), \quad v = \frac{x}{y}$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= f_{xx} + \frac{1}{y} f_{vz}$$

$$\frac{\partial^2 u}{\partial x^2} = f_{xx} + \frac{2}{y} f_{xz} + \frac{1}{y^2} f_{zz}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial y}$$

$$= f_{yy} - f_{vz} + \frac{x}{y^2} f_{vz}$$

$T_4:$

$$(1) \cdot z = \varphi(x^2 + y^2)$$

$$\frac{\partial z}{\partial x} = \varphi'(x^2 + y^2) \cdot 2x$$

$$\frac{\partial z}{\partial y} = \varphi'(x^2 + y^2) \cdot 2y$$

$$y \cdot \frac{\partial z}{\partial x} - x \cdot \frac{\partial z}{\partial y} = 2xy [\varphi'(x^2 + y^2) - \varphi'(x^2 + y^2)] = 0$$

$$(2) \cdot u = y \varphi(x^2 - y^2)$$

$$\frac{\partial u}{\partial x} = y \varphi'(x^2 - y^2) \cdot 2x$$

$$\frac{\partial u}{\partial y} = \varphi(x^2 - y^2) + y \varphi'(x^2 - y^2) \cdot (-2y)$$

$$y \cdot \frac{\partial u}{\partial x} + x \cdot \frac{\partial u}{\partial y} = x \cdot \varphi'(x^2 - y^2) = \frac{x \cdot u}{y} \quad \text{得证}$$

7.2 若 u, v 为 x, y 的函数, $x = r \cos \theta, y = r \sin \theta$, 试由

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cos \theta, \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \sin \theta$$

(3)

$$\frac{\partial u}{\partial x} = \varphi(x+y) + x \frac{d\varphi(x+y)}{dx} + y \cdot \frac{d\varphi(x+y)}{dx}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{d\varphi(x+y)}{dx} + \frac{d\varphi(x+y)}{dx} + x \cdot \frac{d}{dx} \frac{d\varphi(x+y)}{dx} + y \cdot \frac{d}{dx} \frac{d\varphi(x+y)}{dx}$$

$$= 2 \frac{d\varphi(x+y)}{dx} + x \frac{d}{dx} \frac{d\varphi(x+y)}{dx} + y \frac{d}{dx} \frac{d\varphi(x+y)}{dx}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{d\varphi(x+y)}{dy} + x \frac{d}{dy} \frac{d\varphi(x+y)}{dx} + y \frac{d}{dy} \frac{d\varphi(x+y)}{dx}$$

同理得:

$$d\varphi(x+y)$$

F(

同法得:

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{d\varphi(x+y)}{dy} + x \frac{d}{dx} \frac{d\varphi(x+y)}{dy} + \frac{d}{dx} \frac{d\varphi(x+y)}{dx} + y \frac{d}{dy} \frac{d\varphi(x+y)}{dy}.$$

对变

若 F

$$\frac{\partial^2 u}{\partial y^2} = x \frac{d}{dy} \frac{d\varphi(x+y)}{dy} + 2 \frac{d}{dy} \frac{d\varphi(x+y)}{dy} + y \frac{d}{dy} \frac{d\varphi(x+y)}{dy}.$$

上式代入.

$$\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = \left[\frac{d}{dx} \frac{d\varphi(x+y)}{dx} - \frac{d}{dy} \frac{d\varphi(x+y)}{dy} \right] \varphi(x+y).$$

$$\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr}$$

证明

T6: 设 $u = f(r)$ $r = \sqrt{x^2 + y^2}$

$$\frac{\partial u}{\partial x} = f'(r) \cdot \frac{dr}{dx} = f'(r) \cdot \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{x}{x^2 + y^2} f''(r) + f'(r) \cdot \frac{\sqrt{x^2 + y^2} - \frac{x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2}$$

$$= \frac{x^2}{x^2 + y^2} f''(r) + f'(r) \cdot \frac{-\frac{y^2}{\sqrt{x^2 + y^2}}}{(x^2 + y^2)^{\frac{3}{2}}}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{x^2 + y^2} f''(r) + f'(r) \cdot \frac{-\frac{x^2}{\sqrt{x^2 + y^2}}}{(x^2 + y^2)^{\frac{3}{2}}}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r) = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr}$$

具有
满足

F(将
对

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r) = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr}$$

T7:

由 $u = u(x, y)$, $v = v(x, y)$ 连续可微可以知道:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{1}{\cos \theta} + \frac{\partial u}{\partial \theta} \cdot \frac{-1}{r \sin \theta}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{1}{\sin \theta} + \frac{\partial u}{\partial \theta} \cdot \frac{1}{r \cos \theta}$$

$$\therefore \left. \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial u}{\partial x} \end{aligned} \right\}$$

7. 若 u, v 为 x, y 的函数, $x = r \cos \theta, y = r \sin \theta$, 试由

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

明等式

8. 设 $f(x, y)$

这样性质的

设 φ 与

设 $u =$

$$\left\{ \begin{array}{l} \frac{\partial u}{\partial r} \cdot \frac{1}{\cos \theta} + \frac{\partial u}{\partial \theta} \cdot \frac{1}{-r \sin \theta} = \frac{\partial v}{\partial r} \cdot \frac{1}{\sin \theta} + \frac{\partial v}{\partial \theta} \cdot \frac{1}{r \cos \theta} \\ \frac{\partial u}{\partial r} \cdot \frac{1}{\sin \theta} + \frac{\partial u}{\partial \theta} \cdot \frac{1}{r \cos \theta} = \frac{\partial v}{\partial r} \cdot \frac{1}{\cos \theta} + \frac{\partial v}{\partial \theta} \cdot \frac{1}{-r \sin \theta} \end{array} \right.$$

解得:

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

得证.

Tg:

T₉:

$$\frac{\partial z}{\partial x} = \varphi\left(\frac{y}{x}\right) - \frac{y}{x} \varphi'\left(\frac{y}{x}\right) - x \cdot \frac{y}{x^2} \psi'\left(\frac{y}{x}\right)$$

$$\frac{\partial z}{\partial y} = \varphi'\left(\frac{y}{x}\right) + \frac{1}{x} \psi'\left(\frac{y}{x}\right)$$

则 $\frac{\partial^2 z}{\partial x^2} = \frac{y^2}{x^3} \varphi'' + \frac{2y}{x^3} \varphi''' + \frac{y^2}{x^4} \varphi''''$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{y}{x^2} \varphi'' - \frac{1}{x^2} \varphi' - \frac{y}{x^3} \psi''$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{1}{x} \varphi'' + \frac{1}{x} \psi''$$

$$\Rightarrow x^2 \cdot \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0$$

将 F(对 若

T₄:

(5) {

8. 方

9. 设

T₂:

... 对 x 求偏导. 得:

$$f_1' + f_2' \frac{\partial z}{\partial x} + f_3' \left(\frac{\partial z}{\partial x} + 1 \right) = 0.$$

$$\Rightarrow \frac{\partial z}{\partial x} = - \frac{f_1' + f_3'}{f_2' + f_3'}$$

对 y 求偏导. 得:

~~$$f_1' + f_2' \frac{\partial z}{\partial x} + f_3' \frac{\partial z}{\partial x}$$~~

$$f_1' + f_2' \left(1 + \frac{\partial z}{\partial y} \right) + f_3' \frac{\partial z}{\partial y} = 0.$$

$$\Rightarrow \frac{\partial z}{\partial y} = - \frac{f_1' + f_2'}{f_2' + f_3'}$$

(2)

$$f'_1 + f'_3$$

(12)

$$u = xz, \quad v = z - y$$

则:

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\Rightarrow \frac{\partial z}{\partial x} = f_u (z + x \frac{\partial z}{\partial x}) + f_v \frac{\partial z}{\partial x}$$

整理得:

$$\frac{\partial z}{\partial x} = f_u z + x f_u \frac{\partial z}{\partial x} + f_v \frac{\partial z}{\partial x}$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{f_u z}{1 - x f_u - f_v}$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$\Rightarrow \frac{\partial z}{\partial y} = x \frac{\partial z}{\partial y} + \frac{\partial z}{\partial y}$$

整理得:

$$\frac{\partial v}{\partial y} = \frac{\partial z}{\partial y} - 1$$

$$\frac{\partial z}{\partial y} = \frac{-f_v}{1 - x f_u - f_v} = \frac{f_v}{f_v + x f_u - 1}$$

$$\frac{\partial}{\partial x} + \frac{\partial}{\partial y}$$

$$(cx - c$$

$$F_1' + F_2' (1 - \frac{\partial z}{\partial y}) + F_3' \frac{\partial z}{\partial y} = 0.$$

$$= bx$$

$$\text{解: } \frac{\partial z}{\partial y} = \frac{F_1' - F_2'}{F_3' - F_2'}.$$

$$(4) \frac{f_6}{w_6} + \frac{n_6}{p_6} \cdot \frac{f_6}{n_6} = \frac{f_6}{p_6}$$

$$\text{由 } f(x, y, z) = 0 \text{ 得}$$

$$\frac{\partial z}{\partial x} = - \frac{f_1 + f_2 + f_3}{f_3} \quad \frac{\partial z}{\partial y} = - \frac{f_2 + f_3}{f_3}$$

$$\frac{\partial^2 z}{\partial x^2} = - \frac{\partial(\frac{\partial z}{\partial x})}{\partial x} = - \frac{1}{f_3} [f_{11} + f_{12} + (1 + \frac{\partial z}{\partial x}) f_{13} + f_{21} + f_{22} + (1 + \frac{\partial z}{\partial x}) f_{23}]$$

$$\frac{f_1 + f_2}{f_3} [f_{31} + f_{32} + (1 + \frac{\partial z}{\partial x}) f_{33}]$$

T4:

(5) $\begin{cases} u = \\ v = \end{cases}$

对 $ax+by+cz=z$ ($x^2+y^2+z^2$) 两侧取微分, 得:

$$a dx + b dy + c dz = (2x dx + 2y dy + 2z dz)$$

因此可得:

$$[c - 2z \Phi'] dz = (2x \Phi' - a) dx + (2y \Phi' - b) dy$$

$$\text{于是 } \frac{\partial z}{\partial x} = - \frac{2x \Phi' - a}{2z \Phi' - c}$$

$$\frac{\partial z}{\partial y} = - \frac{2y \Phi' - b}{2z \Phi' - c}$$

代入

$$(cy - bz) \frac{\partial z}{\partial x} + (az - cx) \frac{\partial z}{\partial y} = bx - ay \quad \text{中, 得:}$$

$$(cy - bz) \frac{\partial z}{\partial x} + (az - cx) \frac{\partial z}{\partial y} = (bx - ay) \frac{2x \Phi' - a}{2z \Phi' - c}$$

$$+ (cx - az) \frac{2y \Phi' - b}{2z \Phi' - c}$$

整理, 得:

$$cx = bx - ay$$

综上: 可以证明得到 $z = z(x, y)$ 满足方程...

为极坐标方程

10. 设 x

11. 设 r

12. 设:

13. 设:

14. 设

15. 设:

变换为 $\frac{\partial^2 u}{\partial \xi \partial \eta}$

16. 证

$$\frac{\partial \varphi}{\partial v} = \frac{\partial \psi}{\partial u}$$

17. 设

18. 作

变换为 $\frac{\partial^2 u}{\partial \xi \partial \tau}$

16. 证

$$\frac{\partial \varphi}{\partial v} = -\frac{\partial \psi}{\partial u}$$

17. 设

18. 作

(1) 证

$$+ (cx - az) \frac{2y\psi' - b}{2z\psi' - c}$$

整理得:

$$= bx - ay$$

证上: 可以证明得证 $z = z(x, y)$ 满足方程...

TS:

(2) i 对 $\varphi(cx - az, cy - bz) = 0$ 两边对 x 求偏导得:

$$\varphi_1 (c - a \frac{\partial z}{\partial x}) + \varphi_2 (-b \frac{\partial z}{\partial x}) = 0$$

$$\text{即 } \frac{\partial z}{\partial x} = \frac{\varphi_1 c}{\varphi_1 a + \varphi_2 b}$$

$$\varphi(cx - az, cy - bz) = 0$$

(5) 对 y 求偏导得:

8. 方程

9. 设

$$\varphi_1(-a \frac{\partial z}{\partial y}) + \varphi_2(cx - b \frac{\partial z}{\partial y}) = 0$$

$$\Rightarrow \frac{\partial z}{\partial y} = \frac{\varphi_2 c}{\varphi_1 a + \varphi_2 b}$$

$$\Rightarrow a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{\varphi_1 a c}{\varphi_1 a + \varphi_2 b} + \frac{\varphi_2 b c}{\varphi_1 a + \varphi_2 b}$$

$$= c$$

故得证.

为极坐标方

10. 设

11. 设

12. 设

13. 设

14. 设

15. 设

14.
15.

证得证。
= 0

变换为 $\frac{\partial}{\partial u}$

16.

$$F(x+2y^{-1}, y+z x^{-1}) = 0$$

$\frac{\partial \varphi}{\partial u} = -\frac{\partial \varphi}{\partial v}$

17.

$$F_1(x+2y^{-1}, y+z x^{-1})(1+y^{-1}\frac{\partial z}{\partial x}) + F_2(x+2y^{-1}, y+z x^{-1})(-z x^{-2} + x^{-1}\frac{\partial z}{\partial x})$$

18.

(1)

将上面两个方程相加得：

(2)

$$F_1(x+2y^{-1}, y+z x^{-1})(1+y^{-1}\frac{\partial z}{\partial x} - z y^{-2} + y^{-1}\frac{\partial z}{\partial y}) + F_2(x+2y^{-1}, y+z x^{-1})$$

(3)

$$(-z x^{-2} + x^{-1}\frac{\partial z}{\partial x} + 1 + x^{-1}\frac{\partial z}{\partial y}) = 0 \Rightarrow (\frac{56}{x^6} d -) + (\frac{76}{x^6} d -) + p$$

$\Rightarrow$

$$x \frac{\partial z}{\partial x} - z + y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} - z + y \frac{\partial z}{\partial y} = 0$$

$$\frac{1}{x^6} = \frac{56}{x^6} p$$

$$\Rightarrow x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - x y$$

T9: 设 $x = r \cos \theta$, $y = r \sin \theta$ 则

$$\Rightarrow \frac{dx}{dt} = \frac{dr}{dt} \cos \theta - r \sin \theta \frac{d\theta}{dt}$$

$$\frac{dy}{dt} = \frac{dr}{dt} \sin \theta + r \cos \theta \frac{d\theta}{dt}$$

$$\Rightarrow \begin{cases} \frac{dx}{dt} = r \sin \theta + k r^3 \cos \theta \\ \frac{dy}{dt} = -r \cos \theta + k r^3 \sin \theta \end{cases}$$

对此可得:

$$\frac{dr}{dt} = \frac{dx}{dt} \cos \theta + \frac{dy}{dt} \sin \theta = (r \sin \theta + k r^3 \cos \theta) \cos \theta + (-r \cos \theta + k r^3 \sin \theta) \sin \theta = k r^3$$

$$\frac{d\theta}{dt} = \frac{1}{r} \left(-\frac{dx}{dt} \sin \theta + \frac{dy}{dt} \cos \theta \right)$$

$$= \frac{1}{r} [-(r \sin \theta + k r^3 \cos \theta) \sin \theta + (-r \cos \theta + k r^3 \sin \theta) \cos \theta]$$

$$= -1$$

$$\Rightarrow \begin{cases} \frac{dr}{dt} = k r^3 \\ \frac{d\theta}{dt} = -1 \end{cases}$$

即为极坐标方程.